

XH-M609

Operating Instructions

12-36 V DC battery low-voltage disconnect module

Disconnects the load when the battery falls below the cut-off setting. Reconnects after the voltage recovers to the restart threshold and the selected delay expires.

1 Connect the battery and load

Switch off the supply before wiring. Connect the battery to **IN+** / **IN-** and the load to **OUT+** / **OUT-**, following the markings on your board. Fit a suitable fuse near the battery positive terminal. Check polarity and the unit's voltage/current ratings before powering up.

2 Use the buttons

Start from the normal voltage display. In a setting mode, use **+** or **-** to adjust the value.

Button action	Function
Press + once	View the low-voltage cut-off setting.
Press - once	View the hysteresis (restart difference).
Double-click +	Set the cut-off voltage.
Double-click -	Set the hysteresis voltage difference.
Hold +	Calibrate the displayed input voltage.
Hold -	Set the restart delay: 0-10 minutes.

Save: Release the buttons and wait until the normal voltage display returns (about 3 seconds on the documented version).

3 Set the restart threshold

Restart voltage = cut-off voltage + hysteresis

Example only: a 12.0 V cut-off plus 2.0 V hysteresis gives a 14.0 V restart threshold. Hysteresis is a difference, not the complete restart voltage. Choose thresholds specified for your battery chemistry and cell count.

4 Calibration and delay

Calibration: Measure voltage across **IN+** and **IN-** with an accurate multimeter. Hold **+** until the display flashes, then adjust it to match the meter.

Delay: Hold **-** until the setting flashes. Set the wait before reconnection; **000** means no delay and **005** means five minutes.

Version note: These controls apply to the double-click version. Other published XH-M609 instructions use long presses to set cut-off and hysteresis. If your controls differ, check the instructions supplied with your board before changing settings.